DUAL-USE WILDFIRE DIGITAL TWIN

WildfireTwinLA

Prepare before the fire.
Prioritize outreach during it.

A browser-based 3D digital twin that connects mitigation planning, public fire observations, spread simulation and inspectable household outreach.

OBSERVED · MODELLED · REPLAY · SIMULATED CONTACT



BEFORE

Plan where to prepare

Compare fuel breaks, prescribed burns, thinning and home hardening under the same ignition and wind.

UNCERTAINTY STAYS VISIBLE

DURING

Replay and respond

Fuse reported ignitions, VIIRS passes and wind. Rank fictional households by estimated fire arrival and keep unanswered contacts visible.

OFFICIAL ORDERS STAY SEPARATE

A real-data Norte Fire prototype

After-the-fact replay of Oct 16, 2026 in Orange County. The same 12.8 km landscape supports prevention planning and emergency outreach.

512×512
25 m simulation grid

50,408
OpenStreetMap buildings

4
NASA VIIRS passes

6
parallel demo call lines

WHAT POWERS IT

USGS 3DEP terrain · LANDFIRE fuels · Sentinel-2 imagery · OSM buildings and roads · NASA FIRMS hotspots · archived NOAA HRRR wind

EVIDENCE, NOT OVERCLAIMS

55 of 60 satellite pixels were reached by the replay model at overpass time.

43% overlap with the temporary VIIRS-detected area.

The temporary area is coarse and non-official. The model does not include firefighting suppression and still overestimates building impacts.

github.com/ZQC8848/WildfireTwinLA

Qinchuan · Tia · Lucia

RESPOND

PLAN

Experimental, unofficial model outputs. Demo contacts and call outcomes are fictional. Follow official evacuation instructions.